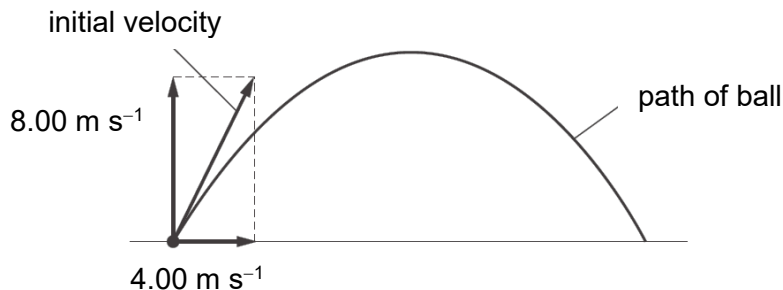


- 3 An astronaut on the Moon, where there is no air resistance, throws a ball. The ball's initial velocity has a vertical component of 8.00 m s^{-1} and a horizontal component of 4.00 m s^{-1} , as shown.



The acceleration of free fall on the Moon is 1.62 m s^{-2} .

What will be the speed of the ball 9.00 s after being thrown?

- | | | | | | | | |
|----------|-------------------------|----------|-------------------------|----------|-------------------------|----------|-------------------------|
| A | 6.60 m s^{-1} | B | 7.70 m s^{-1} | C | 10.6 m s^{-1} | D | 14.6 m s^{-1} |
|----------|-------------------------|----------|-------------------------|----------|-------------------------|----------|-------------------------|

Answer: B

Take upwards and rightwards as positive directions.

Vertical direction:

$$\begin{aligned}
 v_y &= u_y + a_y t \\
 &= 8.00 + (-1.62)(9.00) \\
 &= -6.58 \text{ m s}^{-1}
 \end{aligned}$$

Horizontal direction:

$$\begin{aligned}
 v_x &= u_x + a_x t \\
 &= 4.00 + (0)(9.00) \\
 &= 4.00 \text{ m s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{speed after } 9.00 \text{ s} &= \sqrt{(-6.58)^2 + (4.00)^2} \\
 &= 7.70 \text{ m s}^{-1}
 \end{aligned}$$

- 4 A uniform square metal sheet of length x is cut into an 'L' shape